1. Project Title

Learnifyx

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2. Project Introduction

Learnifyx is a web-based platform designed to manage, deliver, and track online learning activities. It allows educational institutions, training companies, and businesses to create courses, enroll learners, monitor progress, and assess performance from a centralized system.

This project is intended to streamline the complete learning lifecycle, from course creation and student enrollment to lesson delivery, quizzes, certifications, and reporting.

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3. Problem the Project Solves

Many organizations and educational providers struggle with scattered learning materials, manual attendance tracking, inconsistent assessments, and limited visibility into learner progress. An LMS solves these issues by providing:

- Centralized course and content management
- Structured online learning delivery
- Automated enrollment and progress tracking
- Online assessments and grading
- Performance analytics and reporting
- Better communication between instructors and learners

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4. Target Users

- Educational institutions
- Coaching centres and academies
- Corporate training teams
- HR and employee onboarding departments
- Online educators and course creators
- Students, trainees, and employees

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5. Key Features

- User registration and secure login
- Role-based access control
- Course creation and management
- Lesson and module organization
- Video, PDF, and document-based learning content
- Student enrollment management
- Assignments and quiz management
- Progress tracking and completion status
- Certificates for completed courses
- Notifications and announcements
- Reports and dashboard analytics
- Payment integration for paid courses

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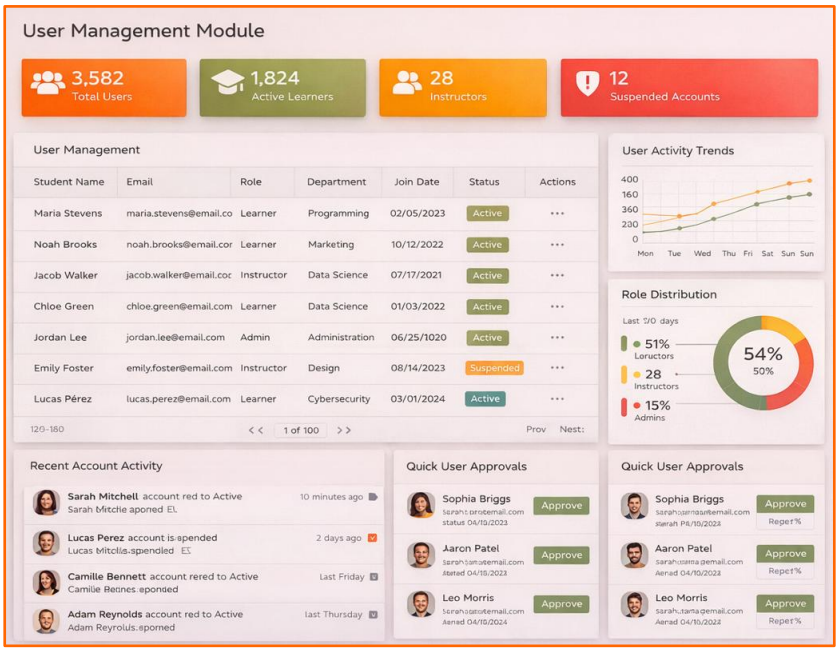
6. System Modules

1. Authentication Module

- User signup, login, logout
- Password reset
- Email verification
- Role-based authentication

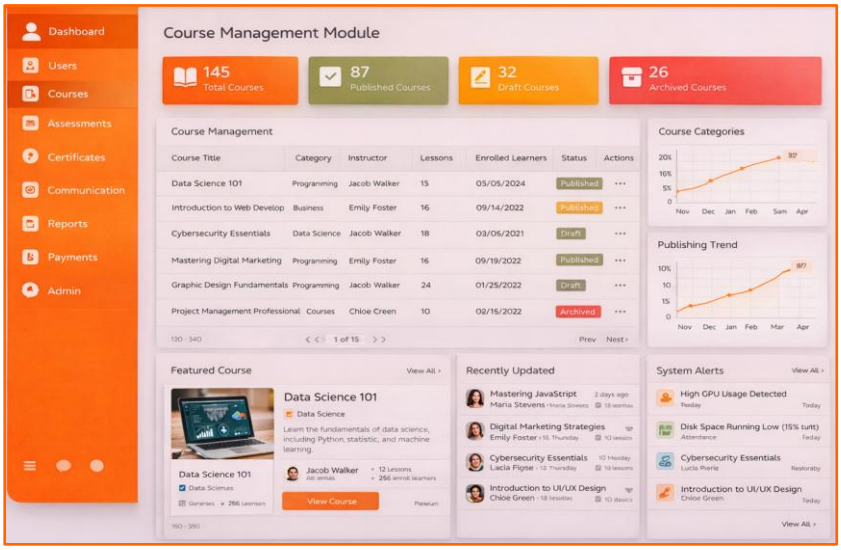
2. User Management Module

- Manage students, instructors, and admins
- Profile management
- Account status control



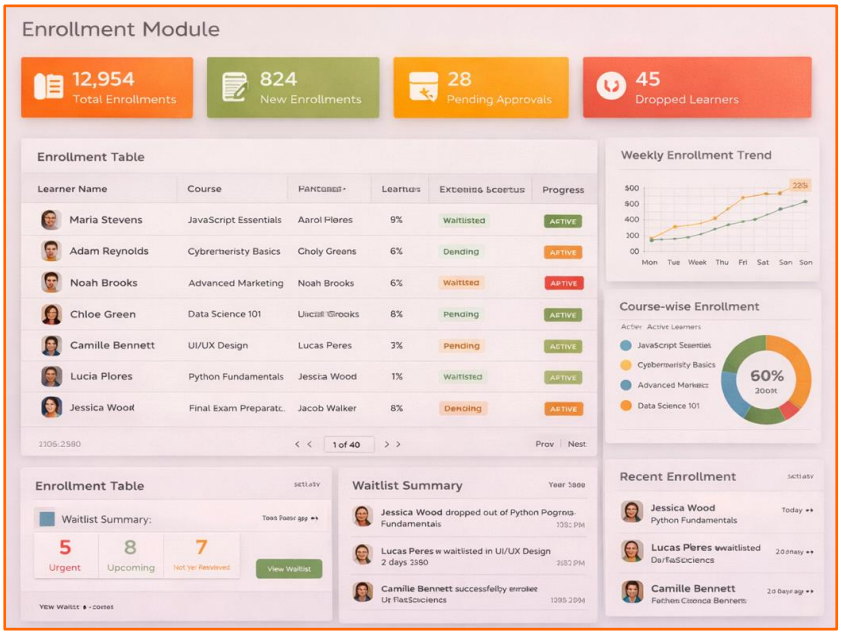
3. Course Management Module

- Create, edit, publish, and archive courses
- Organize courses by category
- Add modules, chapters, and lessons
- Upload multimedia content



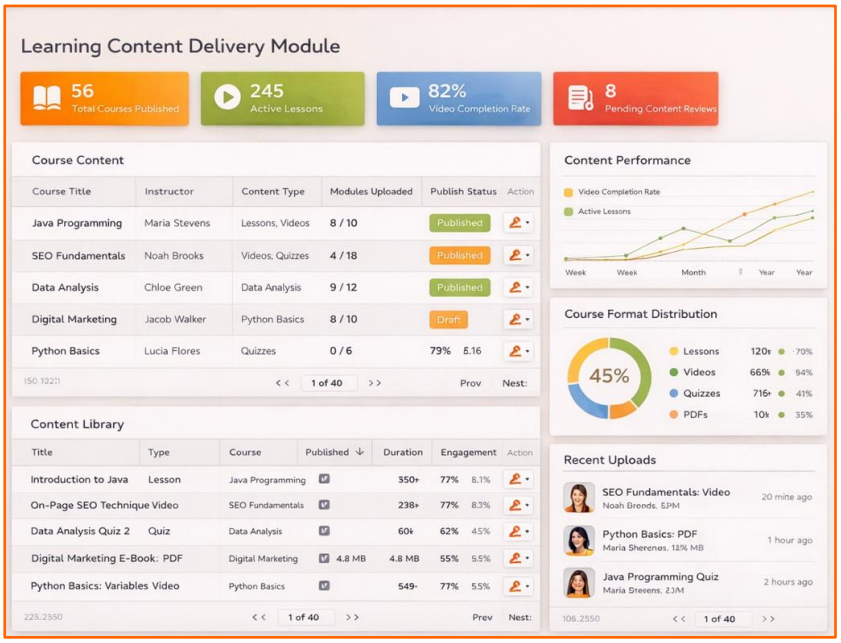
4. Enrollment Module

- Manual and automatic enrollment
- Batch enrollment
- Enrollment history
- Access control per course



5. Learning Content Delivery Module

- Video lessons
- Downloadable resources
- Lesson completion tracking
- Content sequencing



6. Assessment Module

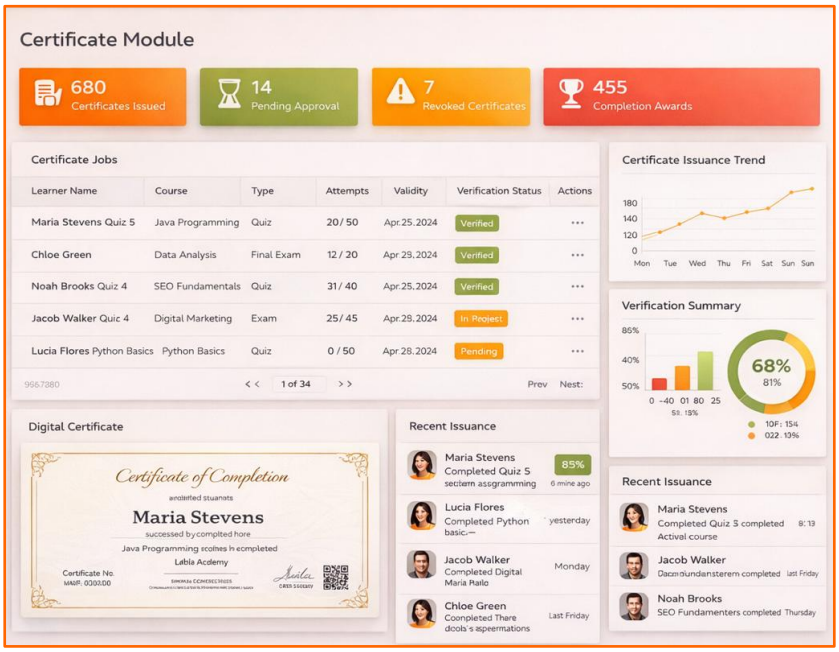
- Quizzes and exams
- Assignments and submissions
- Auto and manual grading
- Result generation

7. Progress Tracking Module

- Course completion percentage
- Quiz and assignment scores
- Learning activity logs
- Attendance or participation tracking

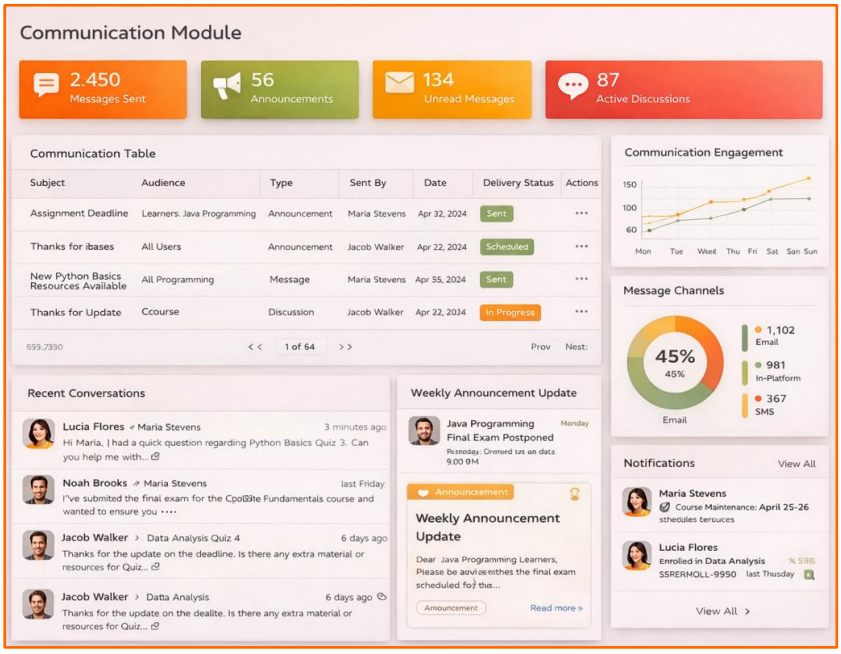
8. Certificate Module

- Auto-generate certificates
- Download and share certificates
- Completion-based eligibility rules



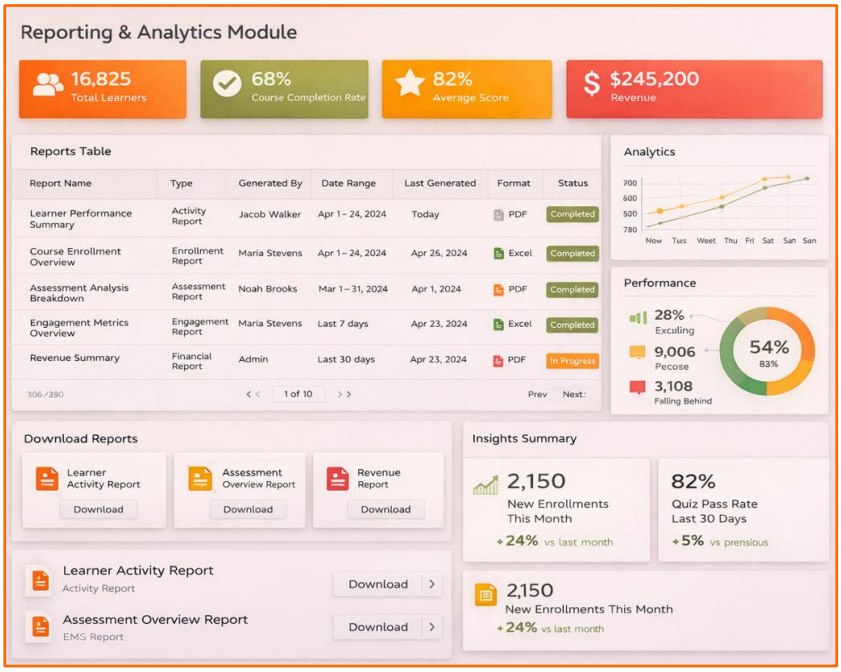
9. Communication Module

- Course announcements
- Email or in-app notifications
- Student-instructor messaging
- Discussion forums or comment threads



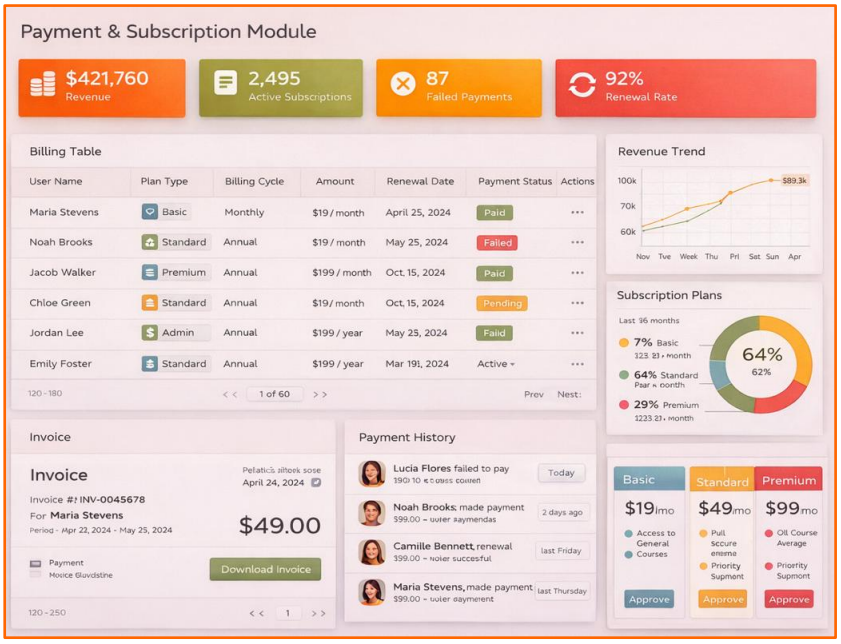
10. Reporting and Analytics Module

- Enrollment reports
- Course performance reports
- User activity reports
- Completion and dropout analysis



11. Payment and Subscription Module

- Paid course access
- Subscription plans
- Payment history
- Coupon or discount support



12. Admin Panel Module

- Dashboard overview
- User and course moderation
- Platform settings
- Reports and system management

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7. User Roles

Admin

- Manage platform users
- Approve or manage instructors
- Create categories
- View overall reports
- Configure system settings
- Monitor platform usage

Instructor / Trainer

- Create and manage courses
- Upload learning materials
- Create quizzes and assignments
- Review submissions
- Track student progress
- Communicate with learners

Student / Learner

- Browse and enroll in courses
- Access lessons and resources
- Attempt quizzes and assignments
- Track learning progress
- Download certificates
- Receive notifications

Core Functional Requirements

- The system must allow users to register and log in securely.
- The system must support multiple user roles with role-based permissions.
- Instructors must be able to create and manage courses.
- Students must be able to enroll in available courses.
- The system must support uploading video, document, and text-based content.
- Learners must be able to mark lessons as completed.
- The system must provide quizzes, assignments, and evaluations.
- Admins must be able to monitor users, courses, and reports.
- The platform must generate progress data and course analytics.
- The system should issue certificates on successful completion.
- The platform should support notifications and announcements.

Admin Panel Features

- Admin dashboard with KPIs
- Total users, instructors, and students count
- Total courses and active enrollments
- Revenue overview for paid plans
- Course approval and moderation
- User suspension and activation
- Category and tag management
- Report exports
- Site settings and branding control
- Banner, email, and notification management

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8. User Workflow / Journey

Student Journey

- User signs up and logs in
- User browses available courses
- User enrolls in a course
- User accesses modules and lessons
- User completes quizzes and assignments
- User tracks progress from dashboard
- User completes the course
- User downloads certificate

Instructor Journey

- Instructor registers or gets approved
- Instructor creates a course
- Instructor uploads lessons and resources
- Instructor publishes the course
- Students enroll in the course
- Instructor monitors learner performance
- Instructor reviews assignments and feedback

Admin Journey

- Admin logs into dashboard
- Reviews system activity and reports
- Manages users and course approvals
- Monitors enrollments, payments, and performance
- Updates settings and platform policies

9. Tech Stack

Backend

- ASP.NET MVC

Frontend

- jQuery
- Bootstrap

Database

- MS SQL Server